Artificial Intelligence

Unit 1 Study Notes with Theory

Introduction to Artificial Intelligence
Foundations of Artificial Intelligence
History of Artificial Intelligence
Intelligent Agents: Agents and Environments
Concept of Rationality
Nature of Environments
Structure of Agents

Reference text: Russell & Norvig, *Artificial Intelligence: A Modern Approach* (4th Edition)

1. Introduction to Artificial Intelligence

1.1 What is AI?

Artificial Intelligence (AI) is one of the fastest-growing fields of Computer Science. It aims to develop machines that can perform tasks requiring human intelligence, such as learning, reasoning, problem-solving, understanding language, perception and decision-making. Humans possess intelligence naturally; AI attempts to replicate these abilities in machines so that they can assist humans in solving complex real-world problems efficiently.

The term has two words. **Artificial** means made by humans (man-made). **Intelligence** means the ability to think, learn, reason, solve problems and make decisions. Therefore, **Artificial Intelligence means human-made intelligence exhibited by machines.** Unlike traditional programs that simply follow predefined instructions, AI systems can learn from data, improve their performance and make intelligent decisions.

What is intelligence?

Intelligence is the ability to learn from experience, acquire and apply knowledge, reason logically, solve problems, understand language, recognize patterns, make decisions and adapt to changing environments.

1.2 Definitions of AI

Who	Definition	Key idea
John McCarthy (1956)	“AI is the science and engineering of making intelligent machines.”	Called the Father of AI; introduced the term at the Dartmouth Conference, 1956.
Russell and Norvig	“AI is the study of intelligent agents that perceive their environment and take actions that maximize their chances of achieving their goals.”	Perceive the environment, process information intelligently, act to achieve goals.
Elaine Rich	“AI is the study of how to make computers do things at which humans are currently better.”	Computers should perform tasks that currently need human intelligence.

1.3 Why do we need AI?

Traditional computer programs	AI systems
Follow fixed instructions Cannot learn from experience Fail when situations change	Learn from experience Improve automatically Handle uncertainty Solve complex real-world problems Assist humans in decision-making

1.4 Goals of AI

- Create machines that can think like humans.
- Enable machines to learn automatically.
- Build systems capable of logical reasoning.
- Develop autonomous decision-making systems.
- Solve complex real-world problems.
- Improve productivity and efficiency.
- Assist humans in dangerous and repetitive tasks.
- Other goals: create expert systems, implement human intelligence in machines, automate repetitive tasks, improve decision-making, learn continuously from experience.

1.5 Capabilities of AI

The capabilities of AI are the abilities that let AI systems perform tasks that normally require human intelligence. They allow machines to learn, reason, understand language, perceive their surroundings, solve problems and make decisions.

Capability	Theory	Methods / Tasks	Example
Learning	Acquiring knowledge from data and improving performance over time without being explicitly programmed.	Supervised, unsupervised, reinforcement learning	ChatGPT improves using training data and user interactions
Reasoning	Drawing logical conclusions from available information, so machines can think logically like humans.	Deductive, inductive, abductive reasoning	Medical expert systems diagnose diseases from symptoms
Problem solving	Finding the best solution among many alternatives by searching possible solutions and selecting the optimal one.	Search algorithms, optimization, heuristics	Google Maps finds the shortest route
Natural Language Processing	Enables computers to understand, interpret and generate human language so humans and machines can communicate naturally.	Translation, chatbots, summarization, question answering, sentiment analysis	ChatGPT, Siri, Alexa, Google Translate
Computer Vision	Enables computers to understand images and videos; it mimics the human visual system.	Object, image and face recognition; medical image analysis	Face Unlock, CCTV surveillance, cancer detection from X-rays
Speech Recognition	Converts spoken language into text or machine commands.	Voice search, voice assistants, smart home devices	Siri, Alexa
Planning	Determining the best sequence of actions to achieve a goal; widely used in robotics and autonomous systems.	Sequence of actions toward a goal	Warehouse robots choose the shortest delivery path
Decision making	Choosing the best action among alternatives, considering data, constraints and objectives.	Evaluating alternatives	Stock prediction, loan approval, medical diagnosis

1.6 Types of Learning

Supervised learning

The model is trained on **labelled data**, where each input (X) is paired with the correct output (Y). In the learning phase the model discovers the relationship between input and output; in the prediction (testing) phase it predicts the output for new input. The goal is to learn a function **f: X to Y** that maps inputs to correct outputs. Example: images of fruits labelled Apple, Banana, Grapes, Orange are used for training; a new strawberry image is then classified.

Unsupervised learning

The data is **unlabelled**: there are no categories or answers. The algorithm analyses the data, measures similarity or distance and discovers patterns or groups, so similar data points are grouped together automatically (clustering). These groups give insights that help in making better decisions.

Reinforcement learning

An **agent** learns by interacting with an **environment**. In each **state** it takes an **action** and receives feedback as a **reward** (good action) or **penalty** (bad action). Its **policy** is the strategy used to choose actions. The goal is to **maximize total reward over time**. Example: a maze agent receives +1 for moving closer to the goal, tries different actions, and after many trials learns the path to the goal.

Term	Meaning
Agent	The learner or decision maker
Environment	Everything outside the agent
State	Current situation of the environment
Action	What the agent can do in that state
Reward / Penalty	Positive / negative feedback
Policy	Strategy used to choose actions
Goal	Maximize total reward over time

1.7 Scope and Applications of AI

Domain	Uses	Example
Healthcare	Disease diagnosis, cancer detection, drug discovery, medical image analysis, robot-assisted surgery	AI detects cancer from MRI and X-ray images
Banking and Finance	Fraud detection, credit card monitoring, credit score prediction, loan approval, risk analysis, virtual assistants	Banks block suspicious transactions
Education	Personalized learning, intelligent tutoring, automatic grading, online learning	Duolingo adapts lessons to student performance
Agriculture	Crop monitoring, disease detection, smart irrigation, yield prediction	Drones monitor crop health
Transportation	Self-driving cars, route optimization, traffic prediction	Google Maps suggests the fastest route
Robotics and Manufacturing	Industrial automation, warehouse and medical robots, quality inspection, predictive maintenance	Robots assemble automobiles
Entertainment	Movie and music recommendation, gaming AI	Netflix recommends movies by user interests
E-commerce	Product recommendation, customer behaviour analysis, dynamic pricing	Amazon suggests products from past purchases
Cyber Security	Spam, malware and intrusion detection	AI filters spam emails
Social Media	Face recognition, friend suggestions, content recommendation	Facebook auto-tags friends in photos

1.8 AI vs Human Intelligence

Aspect	AI	Human intelligence
Origin	Man-made	Natural
Learning	Learns from data	Learns from experience
Speed	Very fast	Comparatively slower
Emotions	No emotions	Has emotions
Creativity	Limited creativity	Highly creative
Memory	Excellent memory	Limited memory
Working time	Works 24x7	Needs rest
Accuracy	High accuracy	Can make mistakes
Intelligence type	Mostly task-specific	General intelligence

1.9 Advantages, Humans' Strengths and Limitations

Advantages of AI over humans	Advantages of humans over AI	Limitations of AI
Processes huge data quickly Works 24x7 without fatigue High accuracy in repetitive tasks Faster data-based decisions Does not forget information easily Performs dangerous jobs safely Reduces human errors	Creativity and innovation Emotional intelligence Ethical and moral reasoning Common sense and intuition Adaptability to completely new situations Social interaction and empathy	Cannot think independently Lacks emotions Limited creativity Depends on training data quality High development and maintenance cost Cannot completely replace human intelligence Raises ethical, privacy and security concerns

1.10 Real-life Examples

AI system	Capability	Application
ChatGPT	NLP, Learning	Education, customer support
Google Maps	Planning, Problem solving	Navigation
Siri	Speech recognition, NLP	Voice assistant
Alexa	NLP	Smart home
Netflix	Learning	Movie recommendation
Amazon	Learning	Product recommendation
Face Unlock	Computer vision	Mobile security
Medical AI	Computer vision	Disease detection
Self-driving cars	Planning, Vision, Decision making	Transportation

2. Foundations of Artificial Intelligence

AI did not develop from a single subject. Many disciplines contributed their ideas, viewpoints, theories and techniques. The history of AI focuses mainly on some important people, events and ideas, although many other contributions were also important. To understand the foundations of AI, we study how different disciplines have tried to answer important questions about thinking, reasoning, learning, decision-making, language and intelligent behaviour.

Discipline	Question it answers
Philosophy	How does the mind work? How can knowledge be used for reasoning?
Mathematics	How can logic, probability and computation be used for intelligent reasoning?
Economics	How can an agent make the best decision to maximise benefit?
Neuroscience	How does the human brain process information?
Psychology	How do humans think, learn, remember and make decisions?
Computer Engineering	How can efficient computers and machines be built to support AI?
Control Theory and Cybernetics	How can systems control their own behaviour to achieve a goal?
Linguistics	How can machines understand and generate human language?

2.1 Philosophy

Philosophy gave the basic ideas about mind, knowledge, reasoning, action, decision making and ethics.

- **Logic and reasoning.** Aristotle (384-322 BCE) developed rules of logic and syllogisms to draw valid conclusions from given premises (All humans are mortal; Socrates is human; therefore Socrates is mortal). Later thinkers such as Leibniz and Hobbes suggested that reasoning could be done by machines. This is the foundation of logical reasoning and rule-based AI systems.
- **Mind and knowledge.** Philosophers debated how the mind works and where knowledge comes from. *Dualism* (Descartes) says mind and matter are different. *Materialism* says the mind is the result of physical processes in the brain. *Empiricism* (Bacon, Locke) says knowledge comes from senses and experience; Hume gave the principle of induction (learning from experience). These ideas are the basis for machine learning and knowledge acquisition.
- **Knowledge to action.** The final aim of intelligence is to take appropriate action. Aristotle explained that actions are justified by connecting goals with knowledge of outcomes, and described a step-by-step process (an early algorithm) for choosing actions. Modern AI uses these ideas in planning, problem solving and intelligent agents.
- **Rational decisions and ethics.** When there are many choices or uncertainty, we should choose the best action. Arnauld and Bernoulli introduced utility and expected utility, the base of modern rational decision making. *Utilitarianism* (Bentham, Mill) says choose the action that gives maximum good (consequentialism). *Deontological ethics* (Kant) says follow moral rules and do the right thing, whatever the outcome.

2.2 Mathematics

Mathematics provides the tools for logical reasoning, dealing with uncertainty, learning from data, solving problems and understanding computation.

- **Formal logic.** George Boole developed Boolean logic (AND, OR, NOT). Gottlob Frege extended it to First-Order Logic (objects and relationships). Logic became an important foundation of early AI. *AI use:* knowledge representation, reasoning and rule-based systems.

- **Probability: reasoning with uncertainty.** Cardano introduced early probability; Pascal and Fermat developed probability methods; Bernoulli and Laplace improved probability and statistical methods; Thomas Bayes introduced **Bayes' Rule** to update probability with new evidence. Example: an AI system may say there is an 80% probability of rain. *AI use:* prediction, diagnosis and decision-making under uncertainty.
- **Statistics: learning from data.** The development of probability and availability of data led to statistics. John Graunt analysed London population data (1662); Ronald Fisher combined probability, experiment design, data analysis and computation. *AI use:* statistics is the foundation of machine learning and data analysis.
- **Godel and the limits of mathematical reasoning.** Kurt Godel's Incompleteness Theorem (1931) showed that in any powerful formal system, some true statements cannot be proved within that system. So not everything can be solved by formal logical rules.
- **NP-completeness.** Cook (1971) and Karp (1972) developed the theory of NP-completeness, which analyses the difficulty (tractability) of problems. NP-complete problems are likely to be very hard to solve efficiently. Many AI problems involve huge numbers of possible solutions and are computationally very difficult.

2.3 Economics

Economics helps an intelligent agent choose the best action among several alternatives with limited resources (time, money, energy) to achieve maximum benefit. **Example:** a self-driving car can take Route A (10 km, heavy traffic, 30 minutes, less fuel) or Route B (14 km, light traffic, 20 minutes, more fuel). Although Route B is longer, it takes less time, so the overall benefit is higher and the car chooses Route B.

2.4 Neuroscience

Neuroscience studies the brain and nervous system. The brain uses billions of interconnected neurons to receive information, process it, learn from experience and produce actions. The flow is: input from senses, neurons carry signals, brain processes information, decision, action. Example: touching a hot object. Skin receptors detect heat, sensory neurons carry the signal, the nervous system recognizes the dangerous stimulus, motor neurons activate muscles, and you pull your hand away. This biological idea inspired **Artificial Neural Networks (ANNs)**: a biological neuron (dendrites, cell body, axon) maps to an artificial neuron (inputs, weights, activation function, output); many artificial neurons form a neural network (input, hidden and output layers); and deep neural networks power tasks such as image recognition.

2.5 Psychology

Psychology is the study of the human mind and behaviour. It explains how humans perceive information, learn from experience, store and recall information, solve problems and make decisions. The cycle is Observe, Think, Learn, Remember, Decide, Act. Example: learning to drive. The learner watches the road and signals, thinks about the situation, learns the rules through practice, remembers past experiences, decides to stop at a red light and presses the brake. These human abilities inspired AI systems that learn from previous data and experience and improve their decisions.

2.6 Computer Engineering

Computer Engineering deals with the design and development of hardware, processors, memory, sensors and other electronic systems. It contributes to AI by providing powerful and efficient hardware that can process large amounts of data and execute AI algorithms quickly. Flow: sensors/input collect data, a processor (CPU/GPU) processes it, an AI/ML algorithm runs, decisions are made, and output or action is produced. AI requires fast processors, memory and storage for large data, sensors to collect information, and GPUs/AI chips to run ML and deep learning efficiently. Example: face recognition in a smartphone. The camera captures the face, the processor handles the image, an AI model identifies facial features, the face is

compared with stored authorized data, and if it matches, the phone unlocks.

2.7 Control Theory and Cybernetics

Control Theory and Cybernetics study how a system can observe its current condition, compare it with a desired goal, and automatically adjust its actions to reach that goal. The most important concept is **feedback**. Steps: set the goal, sense the current condition, compare with the goal, take corrective action, check again and repeat (the feedback loop). Example: an air conditioner with goal 24 degrees C. The sensor detects 30 degrees C, the system compares it with 24, the AC starts cooling, the sensor keeps measuring, and when the room reaches 24 degrees C the AC reduces or stops cooling.

2.8 Linguistics

Linguistics is the scientific study of human language. It helps AI understand how words, grammar, sentence structure, meaning and context are used in communication, and it led to **Natural Language Processing (NLP)**, which enables computers to understand, interpret and generate human language. Flow: human language input, analyse words and grammar, understand meaning and context, process information, generate a human-like response. Example: asking a voice assistant “What is the weather today?” The microphone receives speech, speech recognition converts it to text, language understanding identifies keywords (weather, today), the system fetches weather data, language generation prepares a sentence, and the assistant replies “Today will be sunny.”

3. History of Artificial Intelligence

The history of AI is the story of how ideas about machine intelligence moved from a philosophical question to a practical technology. The key periods are summarised below.

Period	Key events	Examples
1950s	Alan Turing proposes the Turing Test to evaluate machine intelligence	If a machine can imitate a human, it passes the test
1956	The term “Artificial Intelligence” is coined at the Dartmouth Conference	AI becomes an academic field
1980s	Expert systems are introduced	Medical diagnosis programs
2000s to now	Machine Learning and Deep Learning dominate	Self-driving cars, ChatGPT, robotics, image recognition, voice assistants

Year	Milestone	Key person / system
1950	Turing Test: proposed to test whether a machine can exhibit human-like intelligent behaviour	Alan Turing
1956	Term AI coined at the Dartmouth Conference	John McCarthy
1966	First chatbot, ELIZA, developed to simulate human conversation	Joseph Weizenbaum
1980s	Expert Systems (MYCIN) become important AI applications for solving specialized problems	Stanford AI Lab
1997	Deep Blue defeats world chess champion Garry Kasparov	IBM
2011	Watson defeats human champions on the quiz show Jeopardy!	IBM
2016	AlphaGo defeats Go champion Lee Sedol	Google DeepMind
2022-2025	Generative AI (ChatGPT, DALL-E) capable of generating text, images, audio and more	OpenAI

Today, AI can compose music, write poems, drive cars and more. The trend moves from rule-based expert systems (hand-written knowledge) to machine learning (learning from data) to generative AI (creating new content).

4. Intelligent Agents

The aim of this topic is to develop a small set of design principles for building successful agents: systems that can reasonably be called **intelligent**.

4.1 Agents and Environments

An **agent** is anything that can be viewed as perceiving its environment through **sensors** and acting upon that environment through **actuators**. The **environment** is the part of the world that affects what the agent perceives and is also affected by the agent's actions.

Agent	Sensors	Actuators
Human agent	Eyes, ears, other sensory organs	Hands, legs, vocal tract
Robotic agent	Cameras, infrared range finders	Motors and other mechanical devices
Software agent	File contents, network packets, keyboard, mouse, touchscreen, voice	Writing files, sending network packets, displaying information, generating sounds

Percept: the content that an agent's sensors are perceiving at a particular instant.
Percept sequence: the complete history of everything the agent has ever perceived.
An agent's choice of action at any instant can depend on (1) its built-in knowledge and (2) the entire percept sequence observed so far. It cannot depend on something it has not perceived.

Agent function and agent program

Mathematically, an agent's behaviour is described by the **agent function**, which maps a given percept sequence to an action: **Agent function: Percept sequence to Action**. Examples: percept 'room is dirty' gives action Clean/Suck; percept 'room is clean' gives action Move to another room. Theoretically the agent function could be a table of every possible percept sequence and its action, but for most agents that table would be extremely large or infinite unless percept sequences are restricted in length. The **agent program** is the actual implementation of the agent function.

The vacuum-cleaner world

A simple example uses two locations, A and B. Each can be clean or dirty. The agent can move left or right and can suck dirt. The agent function cleans the current square if dirty; otherwise it moves to the other square.

Situation	Percept	Action
A is dirty	[A, Dirty]	Suck
A is clean	[A, Clean]	Right (moves to B)
B is dirty	[B, Dirty]	Suck
B is clean	[B, Clean]	Left

Thus the agent repeatedly perceives the environment, decides what to do and performs an action. The purpose of studying agents is to understand how to design systems capable of nontrivial decision-making and intelligent interaction with their environments.

4.2 Good Behaviour: The Concept of Rationality

A **rational agent** is one that does the right thing. But what does it mean to do the right thing? AI answers using **consequentialism**: an agent's behaviour is evaluated by its consequences. The agent's percepts lead to a sequence of actions, which cause a sequence of environment states. If that sequence of states is desirable, the agent has performed well. Desirability is captured by a **performance measure** that evaluates any given sequence of environment states.

Humans have their own desires and preferences, so rationality for humans means choosing actions that produce desirable states from their own point of view. **Machines** have no desires of their own; the performance measure is (initially) in the mind of the designer or user.

Designing performance measures

It can be hard to formulate a performance measure correctly. A vacuum agent rewarded for the amount of dirt cleaned could clean, dump the dirt, and clean again. A better measure rewards having a clean floor (for example, 1 point per clean square per time step, perhaps with a penalty for electricity and noise). **Rule:** design performance measures according to what we actually want to achieve in the environment, rather than how we think the agent should behave.

What is rational at a given time depends on four things

- The performance measure (criterion of success)
- The agent's prior knowledge of the environment
- The actions the agent can perform
- The agent's percept sequence to date

> **Definition of a rational agent:** For each possible percept sequence, a rational agent should select an action that is expected to maximize its performance measure, given the evidence provided by the percept sequence and whatever built-in knowledge the agent has.

Omniscience, learning and autonomy

- **Rationality is not omniscience.** An omniscient agent knows the actual outcome of its actions, which is impossible in reality. Rationality maximizes *expected* performance, not actual performance. Example: you start crossing the street, and a cargo door falls from an airliner and hits you. You were not irrational, because crossing was reasonable given what you could know.
- **Information gathering.** Rational agents may take actions that modify future percepts. Crossing without looking is not rational (risk too high); looking first gives better information and better expected performance.
- **Learning.** Initial knowledge may be incomplete or wrong, so a rational agent should learn from what it perceives (perceive, learn, improve behaviour). If the environment is fully known and predictable in advance, the agent need not learn. The dung beetle keeps plugging its nest with nonexistent dung when the ball is removed: its built-in assumption fails when violated. The sphex wasp re-checks the burrow and repeats its routine when the caterpillar is moved, never learning that its plan is failing. Agents that cannot learn cannot adapt.
- **Autonomy.** If an agent relies mostly on its designer's prior knowledge rather than its own percepts, it lacks autonomy. A rational agent should be autonomous: it should learn to compensate for partial or incorrect prior knowledge. For example, a learning vacuum agent that predicts where and when dirt will appear performs better.

4.3 The Nature of Environments

To build rational agents we must first understand **task environments**, the “problems” to which rational agents are the “solutions.” The nature of the task environment directly affects the appropriate design of the agent. A task environment is described by **PEAS**: Performance measure, Environment, Actuators, Sensors.

PEAS example: automated taxi driver

Performance measure	Environment	Actuators	Sensors
Safe, fast, legal, comfortable trip, maximize profits, minimize fuel consumption, wear and tear, trip time or cost, violations and disturbances; maximize safety and passenger comfort	Roads (rural, urban, highways), other traffic, pedestrians, police, animals, road works, puddles, potholes, passengers, weather	Steering, accelerator, brake, signal, horn, display, speech	Cameras, radar, lidar, speedometer, GPS, engine sensors, accelerometer, microphones, touchscreen

Other PEAS examples

Agent	Performance	Environment	Actuators	Sensors
Medical diagnosis system	Healthy patient, reduced costs	Patient, hospital, staff	Display questions, tests, diagnoses, treatments	Touchscreen/voice entry of symptoms and findings
Part-picking robot	% of parts in correct bins	Conveyor belt with parts, bins	Jointed arm and hand	Camera, tactile and joint angle sensors
Satellite image analysis	Correct categorization of objects, terrain	Orbiting satellite, downlink, weather	Display of scene categorization	High-resolution digital camera
Refinery controller	Purity, yield, safety	Refinery, raw materials, operators	Valves, pumps, heaters, stirrers, displays	Temperature, pressure, flow, chemical sensors
Interactive English tutor	Student's score on test	Set of students, testing agency	Display exercises, feedback, speech	Keyboard entry, voice

Properties of task environments

Property	Meaning
Fully vs partially observable	*Fully:* sensors give the complete state of the environment. *Partially:* some information is missing or noisy. *Unobservable:* no sensors at all.
Single-agent vs multiagent	*Single:* only one agent affects the environment. *Multiagent:* two or more agents. *Competitive:* conflicting goals (chess). *Cooperative:* common goals (avoiding collisions in taxi driving).
Deterministic vs nondeterministic	*Deterministic:* next state is completely determined by current state and action. *Nondeterministic:* next state is uncertain. If probabilities are known it is *stochastic*; if not quantified, nondeterministic.
Episodic vs sequential	*Episodic:* each episode (percept then action) is independent. *Sequential:* the current action affects future decisions.
Static vs dynamic	*Static:* environment does not change while the agent deliberates. *Dynamic:* it can change. *Semidynamic:* the environment does not change but the performance score does (chess with a clock).
Discrete vs continuous	*Discrete:* finite states, actions and time steps. *Continuous:* infinite values and continuous time, actions and states.
Known vs unknown	*Known:* the agent knows the rules and outcomes (or their probabilities). *Unknown:* the agent must learn how the environment works.

Examples of task environments and their characteristics

Task	Observable	Agents	Deterministic	Episodic / Seq.	Static / Dyn.	Disc. / Cont.
Crossword puzzle	Fully	Single	Deterministic	Sequential	Static	Discrete
Chess with a clock	Fully	Multi	Deterministic	Sequential	Semidynamic	Discrete
Poker	Partially	Multi	Stochastic	Sequential	Static	Discrete
Backgammon	Fully	Multi	Stochastic	Sequential	Static	Discrete

Task	Observable	Agents	Deterministic	Episodic / Seq.	Static / Dyn.	Disc. / Cont.
Taxi driving	Partially	Multi	Stochastic	Sequential	Dynamic	Continuous
Medical diagnosis	Partially	Single	Stochastic	Sequential	Dynamic	Continuous
Image analysis	Fully	Single	Deterministic	Episodic	Semidynamic	Continuous
Part-picking robot	Partially	Single	Stochastic	Episodic	Dynamic	Continuous
Refinery controller	Partially	Single	Stochastic	Sequential	Dynamic	Continuous
English tutor	Partially	Multi	Stochastic	Sequential	Dynamic	Discrete

Three worked examples

Crossword (simple)	Why?	Chess with a clock (medium)	Why?
Fully observable	You see the entire puzzle	Fully observable	You see all pieces on the board
Single agent	Only you are solving it	Multiagent	You and your opponent are both agents
Deterministic	A letter gives a predictable result	Deterministic	A move has a definite result; no dice
Sequential	One answer affects later answers	Sequential	Your move affects later moves
Static	Puzzle does not change while you think	Semidynamic	Board is static but the clock keeps running
Discrete	Letters, boxes and actions are countable	Discrete	Specific squares, pieces and legal moves

Taxi driving (complex): partially observable (cannot know what another driver is thinking or what is hidden behind a vehicle); multiagent (many drivers, pedestrians and road users); stochastic (other drivers behave unpredictably); sequential (what the taxi does now affects what happens later); dynamic (traffic keeps changing while the AI decides); continuous (speed, distance, steering angle, position and time change continuously).

4.4 The Structure of Agents

So far we described agents by their behaviour (the agent function). The job of AI is to design an **agent program** that implements the agent function, the mapping from percepts to actions. The program runs on a computing device with physical sensors and actuators, called the **architecture**.

> **agent = architecture + program**

The program must suit the architecture: if the program suggests actions like Walk, the architecture must have legs. The architecture can be an ordinary PC or a robotic car with computers, cameras and other sensors. In general, the architecture gets percepts from the sensors and gives them to the program, runs the program, and sends the program's chosen actions to the actuators as they are generated.

Agent program vs agent function

Agent program	Agent function
Takes only the **current percept** as input. If it needs past information, it must remember it itself.	Maps **percept sequences** to actions and may depend on the **entire percept history**.

Table-driven agent

The simplest agent program, TABLE-DRIVEN-AGENT, remembers the entire percept sequence and looks it up in a table to find the action.

- A new percept arrives.
- Add it to the end of the percept sequence (memory).
- Look up the whole sequence in the table.
- Get the corresponding action.
- Return that action.

```
function TABLE-DRIVEN-AGENT(percept) returns an action
  persistent: percepts, a sequence, initially empty
              table, a table of actions indexed by percept
              sequences, initially fully specified
  append percept to the end of percepts
  action <- LOOKUP(percepts, table)
  return action
```

Example walk-through: percept A arrives, sequence [A], action Left; percept B arrives, sequence [A, B], action Forward; percept C arrives, sequence [A, B, C], action Stop.

Why the table-driven approach is impractical

Let P be the set of possible percepts and T the lifetime of the agent (total number of percepts). The number of table entries is the sum, for t from 1 to T, of $|P|^t$, which is extremely large. For an automated taxi, visual input from cameras is about 70 MB per second (30 frames/sec, 1080x720 pixels, 24-bit colour), so for one hour the table needs over $10^{600,000,000,000}$ entries. Even for chess, the lookup table has at least 10^{150} entries. Therefore: (a) no physical agent has space to store such a table; (b) the designer cannot spend time creating such a huge table; (c) the agent cannot learn all entries from experience.

> **Key challenge for AI:** produce rational behaviour from a **small program** instead of a vast table, like replacing large tables of square roots with a short program (Newton's method).

Simple reflex vacuum agent

A small program with simple condition-action rules (no huge table), decided from the current percept only.

```
function REFLEX-VACUUM-AGENT([location, status]) returns an action
  if status = Dirty then return Suck
  else if location = A then return Right
  else if location = B then return Left
```

Four basic kinds of agent programs

Almost all intelligent systems are built using one of these four kinds of agent programs:

#	Agent type
1	Simple reflex agents
2	Model-based reflex agents
3	Goal-based agents
4	Utility-based agents

All of these can be converted into **learning agents**, which improve their performance from experience.

Quick Revision

- AI = human-made intelligence in machines; McCarthy coined the term (Dartmouth, 1956).
- Three types of learning: supervised (labelled data, f: X to Y), unsupervised (finds groups), reinforcement (reward and penalty, maximize total reward).
- Eight disciplines: Philosophy, Mathematics, Economics, Neuroscience, Psychology, Computer Engineering, Control Theory and Cybernetics, Linguistics.
- Timeline: Turing 1950, Dartmouth 1956, ELIZA 1966, Expert Systems 1980s, Deep Blue 1997, Watson 2011, AlphaGo 2016, Generative AI 2022 onward.
- Agent = sensors + actuators. Percept = one input; percept sequence = full history; agent function = abstract mapping; agent program = implementation; agent = architecture + program.
- Rational agent maximizes *expected* performance; rationality is not omniscience. Depends on performance measure, prior knowledge, actions and percept sequence.
- PEAS describes a task environment. Seven properties: observable, agents, deterministic, episodic, static, discrete, known.
- Table-driven agents are impractical (huge tables); simple reflex agents use condition-action rules. Four kinds: simple reflex, model-based reflex, goal-based, utility-based.